Hyperspectral Image Classification Based on Multiscale Cross-Branch Response and Second-Order Channel Attention

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***Abstract*— Recently, most convolutional neural network-based methods use convolutional kernels of fixed size to extract features, which ignore the inherent spatial structure information of ground objects and lose spatial details. In addition, rough first-order statistics is not enough to capture subtle differences between different categories and extract nonlocal context information. To address these issues, a hyperspectral image (HSI) classification method based on multiscale cross-branch response and second-order channel attention (MCRSCA) is proposed in this article. First, a multiscale cross-branch response (MCBR) module is proposed, which uses convolution kernels of different sizes for feature extraction. It adds and concatenates the features of different scales, respectively, to obtain rich and complementary spatial context information. Then, element multiplication and element addition are performed on the fused multiscale features to promote the propagation of the multiscale information and enhance the nonlinear expression ability. Next, the second-order channel attention (SOCA) module is designed to interact the channel information through the feature covariance matrix to obtain the long-term dependence between channels. This module pays more attention to the significant channels and suppresses the redundant channels. Finally, the residual connection is used to embed MCBR and SOCA into the residual block to improve the gradient back propagation and accelerate the training process. Experiments on four commonly used HSI benchmark datasets show that the results of MCRSCA are competitive compared with other state-of-the-art methods.**

***Index Terms*— Channel attention, convolutional neural network (CNN), hyperspectral image classification (HSIC), multiscale cross-branch response (MCBR), second-order statistics.**

I. Introduction

With the development of the earth observation technology of remote sensing, the demand for obtaining surface information from high altitudes through sensors is increasing. Hyperspectral image (HSI), as a kind of remote sensing image, contains rich spatial and spectral information [1]. Making full use of this information will help researchers to better identify terrain and analyze the distribution of ground objects. HSI is a kind of cube structure data. Each pixel represents a certain spatial range in the actual scene and contains hundreds of continuous bands to provide detailed spectral information [2]. The abundant information in HSI provides convenience for ground object analysis and attracts a large number of scholars' attention. It also plays an important role in agricultural guidance [3], natural disaster prediction [4], target detection [5], environmental monitoring [6], and other fields. HSI classification (HSIC), as an important task in remote sensing analysis, aims to use limited labeled pixels to assign correct labels to each pixel, so as to support the development of other applications.

In the past decades, many effective traditional methods have been proposed in HSIC. Among these methods, support vector machine (SVM) uses kernel function to map features to high-dimensional space, which can obtain robust classification results, and has been widely used in HSIC [7], [8]. In addition, other methods, such as logistic regression [9], extreme learning machine [10], sparse representation [11], and decision tree [12], were applied to HSIC. In the above methods, each pixel was regarded as a high-dimensional vector. However, there are often redundant components in hundreds of bands in HSI [13], which is easy to cause Hughes phenomenon when the size of labeled samples is small and the spectral dimension is high [14]. Therefore, principal component analysis (PCA) [15] and independent component analysis (ICA) [16] were used to reduce the spectrum dimension and improved the classification accuracy. There is spectral variability in HSI, and using only spectral information may lead to misclassification. Therefore, many methods further improved the classification performance by combining spatial information. The local binary pattern (LBP) [17] was used to extract rich local texture features in HSI, and Gabor filter [18], [19] was used to obtain band-related responses in different directions and different scales from HSI. Luo *et al.* [20] embedded the original data into the low dimensional space and used the intraclass and interclass neighborhood structure graphs to enhance the association between the same classes and reduce the coupling between different classes. Duan *et al.* [21] constructed a nonlinear geodesic sparse hypergraph to obtain the complex relationship between pixel pairs. There are many

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homogeneous spatial structures in HSI. Superpixel segmentation was adopted to maintain the spatial structure information of HSI [22]. However, these traditional methods rely on expert knowledge and need to design feature extractors manually. The feature extraction process and parameter setting are relatively cumbersome.

Algorithm 1 General Procedure of Training and Testing

Input: $X \in R^{W\times H\times B}$: Hyperspectral image; C: the number of classes; s: spatial size; lr: learning rate; T: training epochs;

Output: C_{map}: classification map;

1: Obtain a set of image patches X_l with a spatial size of $s \times$ s for each labeled pixel in X around its spatial neighborhood and a set of corresponding labels Y_l;

2: Divide X_l and Y_l into training set X_{train}, Y_{train} and test set X_{test}, Y_{test};

3: Initialize the weight parameters of the proposed model;

4: for $t = 1$ to T **do**

5: Input the training set X_{train} and Y_{train} into the model;

6: Use Eq. (2) to obtain the features S_1, S_2, S_3 of each stage in turn;

7: Use Eq. (3) to get the predicted labels during training;

8: Use Eq. (4) to get the cross-entropy loss and use the Adam optimizer to update the parameters in the network;

9: end for

10: Get the prediction result from the trained model with the test set X_{test}, Y_{test}, and evaluate the model;

11: Obtain the labels of each pixel and generate a classification map C_{map} from the trained model with the input X_l;

12: Return classification map *Cmap*.

On the contrary, the feature extraction process based on deep learning does not require too much prior knowledge and can often obtain deeper and more abstract features. Chen *et al.* [23] applied stacked autoencoder (SAE) to HSIC and used unsupervised training to obtain deep-level features. Subsequently, the deep belief network (DBN) was used for the first time to extract features through unsupervised pretraining and supervised fine-tuning [24], which proved the potential of deep learning in HSIC. However, the above two methods both stretch the pixels in the spatial neighborhood into vectors, destroying the spatial structure of HSI. Convolutional neural network (CNN) can obtain the local spatial features by sliding the convolution kernel on the image and uses local connections and weight sharing to reduce the trainable parameters. Therefore, CNN is widely used in HSIC [25], [26]. There is a certain correlation between different bands in HSI, so each pixel can be regarded as sequence data. Based on this, Mou *et al.* [27] improved the classification performance with the excellent sequence modeling capabilities of recurrent neural network (RNN). Subsequently, a series of improved methods based on RNN further improved the classification performance [28], [29]. However, when using RNN, the spatial structure information cannot be fully utilized. Hu *et al.* [30] proposed a convolutional long- and short-term memory (ConvLSTM) network, which jointly utilizes the advantages of CNN and long- and short-term memory (LSTM) network in spatial feature extraction and sequence modeling.

In the abovementioned deep learning methods, due to the advantages of CNN in extracting spatial and spectral features, researchers have successively proposed a large number of CNN-based methods. Hu *et al.* [25] proposed 1-D CNN, which used each pixel in HSI as a spectral vector and trained it through a five-layer network. In order to make full use of spatial context information, Makantasis *et al.* [33] proposed 2-D CNN, which used CNN and multilayer perceptron (MLP) to construct spatial and spectral high-level features. Li *et al.* [34] proposed a method based on pixel pairs [CNN-pixel-pair features (PPF)], which increased the number of training samples by calculating the similarity between pixel pairs. With the help of the 3-D structure of HSI, the 3-D CNN method was proposed in [35] and [36], which took all the bands of the image patches as an input and extracted robust spatial and spectral features. However, 3-D convolution increased computational complexity and training time. In order to balance classification performance and calculation time, more abundant spectral-spatial features can be extracted by combining 2-D CNN and 3-D CNN, and time consumption can be reduced [37], [38]. With the deepening of the network, it may cause gradient disappearance and overfitting phenomenon. Therefore, Zhong *et al.* [39] proposed spectral spatial residual network (SSRN) with the help of residual learning, which combined residual learning with spectral and spatial feature learning. Song *et al.* [40] proposed the deep feature fusion network (DFFN). By introducing the residual connection, the residual blocks of multiple stages were fused to learn features of different levels and obtained rich complementary information. In addition, the methods [41], [42] based on fully convolution network (FCN) were applied to HSIC. These methods directly input the whole HSI into the network without additional preprocessing, which reduced the computational burden.

The spatial structure of ground objects in real scenes is not fixed, and the features of a single scale cannot reflect the real distribution of ground objects, so multiscale information is very important. The inception module is proposed in GoogLeNet [43], using 1×1, 3×3, and 5×5 convolution kernels and max-pooling to extract multiscale information. Lee and Kwon [44] proposed context CNN (CDCNN), which placed the convolutional kernel group of multiple scales into the first layer to extract the features of different scales, respectively. The features of multiple scales were concatenated as the input of the subsequent network. Because multiscale features were only learned at a single layer of the network, global multiscale features cannot be learned. Zhang *et al.* [45] proposed a multiscale dense network (MSDN), which extracted features from coarse and fine levels, and made full use of deep and multiscale features in both horizontal and vertical directions. Xie *et al.* [46] proposed a multi-input model, in which patches of different spatial sizes were input into different branches to extract features. Wang *et al.* [47] designed the spectral multiscale subnet and the spatial

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multiscale subnet, respectively, extracted the spectral features under different groups through LSTM, and obtained rich spatial sequence information through ConvLSTM. Gao *et al.* [48] proposed a multiscale residual network (MSRN) with mixed depth-wise convolution, which can obtain lightweight multiscale features by mixing convolution kernels of different sizes in one convolution.

Although the above methods have achieved good results by using spectral and spatial information, there is still room for improvement. First, HSI has the characteristics of limited labeled samples and high spectral dimension, and the methods based on deep learning are often data-driven. How to design a network to improve the classification performance with limited training samples is an urgent problem to be solved. Second, most methods based on CNN use layer by layer stacking to obtain higher level features. However, multiscale information is lost with the deepening of the network. Although some methods used multiscale information, the global multiscale information cannot be obtained only through the simple fusion of multiscale features. In addition, most CNN-based methods only used first-order statistics to extract features. When the spatial and spectral information between different categories in HSI is similar, the traditional first-order network cannot extract the detail information well, which is often easy to cause misclassification. He *et al.* [49] captured the second-order relationship between the spectra in the local window by extracting the covariance map of multiple scale image blocks at the beginning of the network. Zheng *et al.* [50] used covariance pooling at the end of CNN to extract the second-order information of spectral and spatial features. However, the above two methods only extract the second-order information at the beginning or end of the network and cannot fully extract the global second-order information.

In order to solve the above problems, a classification method based on multiscale cross-branch response and second-order channel attention (MCRSCA) is proposed in this article. First of all, the multiscale cross-branch response (MCBR) module is proposed, which extracts multiscale information by using convolution kernels of different sizes and combines feature addition and concatenation to obtain more discriminative features. Cross-branch element addition and multiplication are used to obtain cross-branch first-order and second-order multiscale information. This module guides the model to interactively update parameters according to the information of different branches. After that, the rich multiscale information is fused to enhance the global flow of multiscale information in the network. In addition, the second-order channel attention (SOCA) module is proposed, which applies second-order information to multiple stages of feature extraction and introduces covariance as the second-order representation of features, thereby obtaining rich detail features that are beneficial for classification. By combining the local feature extraction capabilities of CNN with the nonlocal dependence of covariance, complementary global channel-related information and local spatial feature information are obtained. Each item of the covariance matrix represents a pair of channel interactions in the neighborhood space. The average correlation between each channel and all channels is used to capture the long-term context dependence between channels, so as to obtain discriminatory features and further improve the classification performance. Finally, by embedding the second-order information into the attention module, the network can focus on the channels that contribute more to the network.

The main contributions of this article are summarized below.

1) The MCBR is proposed, which integrates feature addition and concatenation to fuse multiscale features to obtain more refined multiscale features and conduct cross-branch interaction of the fused features to promote the global spread of multiscale information in the network.
2) The SOCA captures the correlation between different channels in a local window, fully extracts the second-order statistical information of the features, and enhances the nonlinear modeling ability of the network. In addition, by embedding the second-order information in the attention mechanism module, the SOCA enables the network to focus on channels that contribute more to classification and suppresses redundant channels.
3) The MCBR and the SOCA are merged into the residual block through skip connection, which promotes the flow of multiscale information and second-order information in the whole model. By stacking multiple residual blocks in sequence to form a multistage feature extraction to extract low-level, intermediate-level, and high-level features, respectively, so as to obtain more discriminative features.
4) Experiments are conducted on four commonly used hyperspectral datasets. Even the network is trained with a small amount of training samples, the proposed method also obtains good classification performance, which proves the effectiveness of the proposed method.

The main structure of this article is in the following. Section II introduces the proposed MCRSCA in detail. The experimental results and analysis are illustrated in Section III. Section V concludes this article.

II. Proposed Method

A. Overall Architecture of the MCRSCA

In order to enhance the global flow of multiscale information in the network and obtain more refined multiscale features, as well as obtain the long-term dependence of the channels from the second-order statistics of the features and enhance the nonlinear expression ability of the network, this article proposes MCRSCA. The MCRSCA is mainly composed of three feature extraction stages. The proposed MCBR and SOCA are sequentially stacked in the residual block of each feature extraction stage. The MCBR is used to obtain cross-branch multiscale spatial context information at each stage of the interactive network, and the SOCA is used to capture the correlation between channels and enhance the feature channels that have a large contribution to classification. The overall network architecture is shown in Fig. 1.

As can be seen from Fig. 1, the MCRSCA is divided into three stages, namely, Stage 1, Stage 2, and Stage 3. By sequentially stacking these three stages for feature

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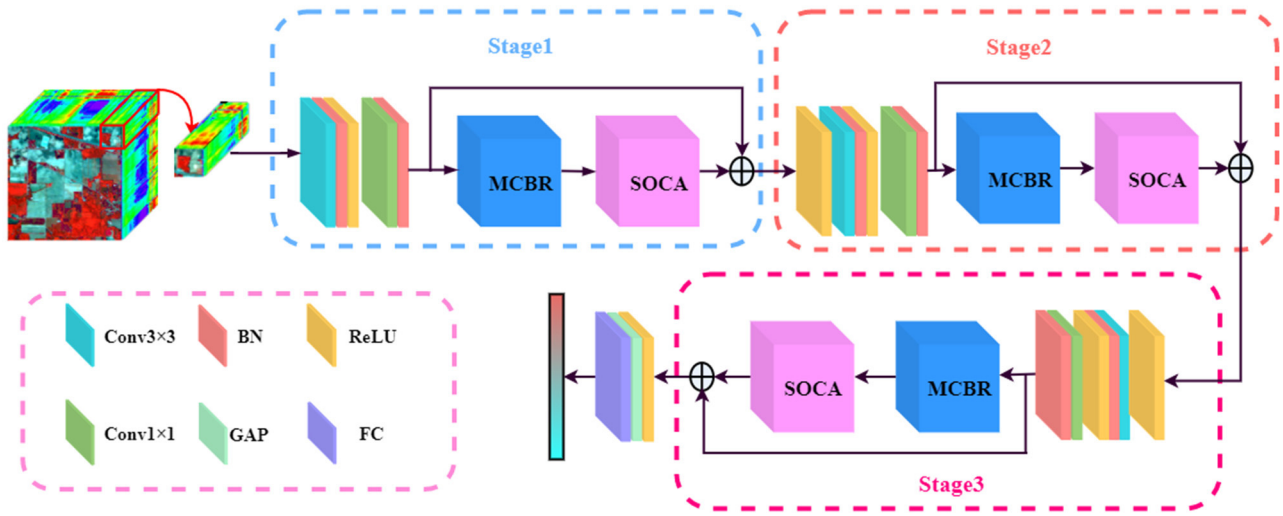


Fig. 1. Overall network architecture of the MCRSCA.

extraction, rich low-level features, middle-level features, and high-level features are sequentially obtained. Low-level features often have a relatively large spatial range to capture more detailed location information, high-level features have strong semantic information, and different levels of information have relevance and complementarity. By combining the features of multiple stages through residual connection, detailed and abstract features can be obtained.

The beginning of each stage is used to extract spatial features. Since the pooling layer may lose some valuable spatial details, 3×3 convolution kernels are used to replace the pooling layer to extract spatial features and reduce the spatial size. Batch normalization (BN) [51] is used to accelerate the convergence speed of network training, and rectified linear units (ReLU) [52] is used to carry out nonlinear transformation of extracted features. After that, 1×1 convolution and BN are used to enhance the nonlinearity as much as possible without significantly increasing the number of parameters.

The labeled samples for HSI are limited. As the network deepens, the network training process is difficult and easily leads to overfitting. In order to solve this problem, skip connections are introduced in each stage, and the gradient back propagation process is improved through residual learning and identity mapping. By embedding the MCBR and the SOCA into the residual block, the network can learn richer multiscale and second-order features. Assuming that the input of the residual block of the ith stage is expressed as I_i, the residual block of the ith stage can be expressed as follows:

$$F_i(I_i) = \sigma(I_i + R(I_i)) \tag{1}$$

where $R(\cdot)$ represents the residual function, and the residual can be obtained after the input I_i is subjected to the feature mapping of the MCBR and the SOCA embedded in the residual block. Here, σ represents the nonlinear activation function ReLU. The identity mapping and the residual are added element-by-element and subjected to nonlinear activation to obtain the output $F_i(I_i)$ of the residual block at the ith stage.

Different stages in the network are used to extract the features of different levels. Assuming that the input of the ith stage is X_i, the features extracted in the ith stage can be formulated as follows:

$$S_i(X_i) = \sigma\left(F_i\left(\sigma\left(X_i * W_{3,i}\right) * W_{1,i}\right)\right) \tag{2}$$

where $*$ represents the convolution operation, and $W_{3,i}$ and $W_{1,i}$, respectively, represent the weights corresponding to the 3×3 convolution kernel and 1×1 convolution kernel in the ith stage. For the simplicity of formula expression, the bias and BN are omitted from the formula. Taking the IP dataset with the input size $15 \times 15 \times 200$ as an example, the final features dimension becomes $9 \times 9 \times 128$ after three stages of feature extraction. Finally, the features are used to the final classification, and the formula is as follows:

$$\widehat{Y}_{\text{train}} = \text{Softmax}(\text{FC}(\text{GAP}(S_3(S_2(S_1(X_{\text{train}})))))) \tag{3}$$

where S_1, S_2, and S_3 represent the features of the first to third stages, respectively. After feature extraction in these three stages, the features are aggregated into a vector with a dimension of 128 by using global average pooling (GAP) [53]. Finally, this vector is input into the full connection (FC) layer, and the probability that the central pixel belongs to each class is obtained through the Softmax activation function.

B. Multiscale Cross-Branch Response Module

Pixels that are adjacent to each other usually have similar spectral characteristics and have a greater probability of belonging to the same category. The combined use of spectral and spatial information will improve the classification performance. However, when the central pixel is located in a heterogeneous area, the direct use of single-scale features may be harmful to classification. The MCBR is shown in Fig. 2.

It can be seen from Fig. 2 that three parallel branches are first designed in the MCBR, and convolution kernels of different scales are applied to these three branches, and BN and ReLU are used after convolution. The size of the convolution

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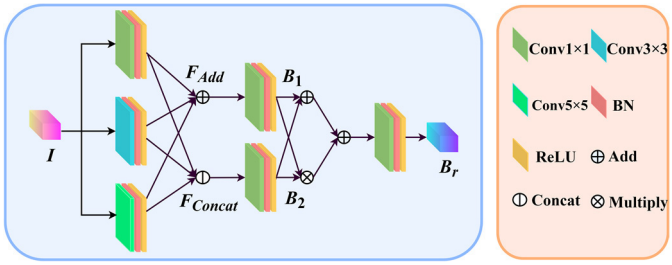


Fig. 2. Architecture of the MCBR.

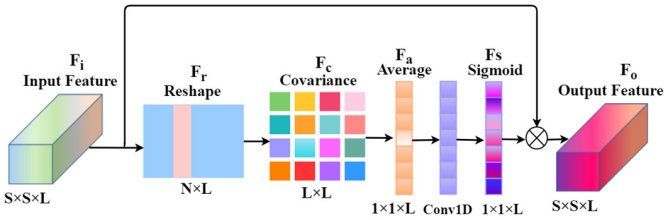


Fig. 3. Network architecture of the SOCA.

kernel on each branch is 1 × 1, 3 × 3, and 5 × 5; 3 × 3 and 5 × 5 convolution kernels are used to extract spatial information of different scales, and 1 × 1 convolution kernels are used to obtain channel correlation and extract spectral information. The combined use of these convolution kernels with different sizes can extract rich spectral and spatial joint information. Next, the feature fusion operation is carried out. Unlike most feature fusions, most of them only use one of the feature elementwise concatenation or elementwise addition. Direct addition may make the elements with larger values dominate the elements with smaller values, thus losing the detailed information, and the direct concatenation may introduce too much redundant information. This article combines elementwise addition and elementwise concatenation to obtain two fused features. The formula is written as follows:

$$F_{\text{Add}}(I) = \sigma(I * W_1) + \sigma(I * W_2) + \sigma(I * W_3) \quad (4)$$

$$F_{\text{Concat}}(I) = [[\sigma(I * W_1), \sigma(I * W_2), \sigma(I * W_3)]] \quad (5)$$

where I represents the input image patches, W_1, W_2, and W_3 represent the weights of the convolution kernels of 1 × 1, 3 × 3, and 5 × 5, respectively. $[[\cdot]]$ represents the feature concatenation operation, F_{Add} represents the multiscale fusion feature after elementwise addition, and F_{Concat} represents the multiscale fusion feature after elementwise concatenation. By making full use of these two multiscale features, the network will obtain richer complementary information.

After obtaining the two fused features, in order to further promote the flow of multiscale information in the network, two branches are used to interact with the multiscale features. Since the channel dimension of F_{Concat} increases through feature concatenation, 1 × 1 convolution is used to reduce the channel dimension of F_{Concat} and keep it consistent with the channel dimension of F_{Add}, and the responses of the two branches are obtained, respectively. The formula is as follows:

$$B_1(I) = \sigma\left(F_{\text{Add}}(I) * W_{1,1}\right) \quad (6)$$

$$B_2(I) = \sigma\left(F_{\text{Concat}}(I) * W_{2,1}\right) \quad (7)$$

where B_1 represents the response of the first branch, $W_{1,1}$ represents the convolution kernel of size 1 × 1 on the first branch, B_2 represents the response of the second branch, and $W_{2,1}$ represents the 1 × 1 convolution kernel of the second branch.

Taking B_1 and B_2 as the input of the two parallel branches, the elementwise multiplication and elementwise addition across the branches are performed on the two branches to obtain the first-order and second-order responses of the feature, then merge the responses of the two branches by linear addition, and finally, perform feature fusion by 1 × 1 convolution. The formula can be written as

$$B_r(I) = \sigma\left((B_1(I) \oplus B_2(I) + B_1(I) \otimes B_2(I)) * W_{r,1}\right) \quad (8)$$

where B_r represents the response after fusing the outputs of the two branches, $\oplus$ and $\otimes$ represent the elementwise addition and elementwise multiplication operators, respectively, and $W_{r,1}$ represents 1 × 1 convolution.

By embedding the MCBR in the residual block and applying MCBR in each stage, multiscale information flows globally in the entire network. The first-order and second-order information across branches are used for interaction, so that each branch can be based on the guidance of another branch to update the weights and enhance the nonlinear modeling ability of the model.

C. Second-Order Channel Attention Module

The importance of bands of HSI to classification is different. Some bands have discrimination, which is conducive to distinguish the differences between different classes. However, some spectral bands are redundant and even interfere with the classification results. Therefore, this article proposed the SOCA to enhance the channels beneficial to classification and suppress the redundant channel information. The network architecture of the SOCA is shown in Fig. 3.

Unlike most attention mechanisms that use GAP to capture the first-order information of features, the SOCA proposed in this article fully considers the correlation between different channels in the local region. The SOCA is embedded in each feature extraction stage to obtain the global second-order context information of the whole model, so as to obtain more detailed features and enhance the nonlinear modeling ability of the model.

Given feature $F_i \in R^{S\times S\times L}$ as the input of the SOCA, where S and L represent the spatial size and the number of channels of the input feature. Then, F_i is reshaped into $F_r \in R^{N\times L}$, where $N = S \times S$ represents N pixels in the input feature, and each sample represents a vector of dimension L. Then, calculate the covariance matrix of the feature to capture the correlation between the channels and use it as the second-order representation of the feature. The formula is expressed as

$$F_c = \frac{1}{N-1}\sum_{i=1}^{N}\left(F_r^i - \overline{F_r}\right)\left(F_r^i - \overline{F_r}\right)^T \in R^{L\times L} \quad (9)$$

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where $F_r^i \in R^L$ represents the ith pixel sample, and $\overline{F_r} \in R^L$ represents the mean vector of N pixel samples. Any element $F_c(i, j)$ in the covariance matrix represents the correlation between the ith channel and the jth channel in the hyperspectral neighborhood space, thereby simultaneously integrating spatial and spectral information. The larger the value of the element in the covariance matrix, the greater the correlation between the features of the corresponding position. Covariance matrix can make full use of the context information of features and capture the pairwise long-range dependences. The elements F_c^i in any row represent the interdependence between the ith channel and all other channels. Therefore, the covariance can be used to obtain the paired second-order statistical information of each channel and all channels, thereby obtaining global channel context information. Then, the average correlation between each channel and all channels is obtained by averaging each row element of the covariance matrix. The formula is as follows:

$$F_a = \frac{1}{L}\sum_{j=1}^{L} F_c^j \in R^{1\times 1\times L}. \tag{10}$$

Unlike most attention mechanisms that use GAP, GAP only obtains the average value of a single channel in the local window, and F_a represents the average correlation dependence between each channel in the local window and all other channels, including global channel context information.

F_a represents the average correlation dependence between channels, and its value is uncertain and cannot reflect the importance of each channel. Therefore, a convolution layer is used to realize local cross-channel information interaction, and the importance of each channel is recalibrated according to the average correlation dependence and the convolution kernel weight. In this article, a 1-D convolution with kernel size k is used to learn the weight to update the importance of each channel. The sigmoid activation function is used to recalibrate the value range of each channel to between 0 and 1 to obtain the weight vector, which makes the model pay attention to the more important channels and suppress the channels that interfere with the classification. The formula is as follows:

$$F_s = \text{sigmoid}(W_k * F_a) \in R^{1\times 1\times L} \tag{11}$$

where W_k represents the weight corresponding to the 1-D convolution kernel with size k. Finally, the output feature map is obtained by multiplying the input feature by the calibrated weight vector. The formula is as follows:

$$F_o = F_i \otimes F_s. \tag{12}$$

D. Training and Test Process

First obtain all the labeled pixels in the HSI, use these labeled pixels as the center to form the image patches set X_l with the surrounding $s \times s$ neighborhood pixels, and obtain the corresponding label set Y_l. Then, according to Y_l, the training set X_{train} and Y_{train} and the test set samples X_{test} and Y_{test} are randomly divided according to a certain proportion. Then, the training set is input into the network, and the spatial spectral features are obtained through the three stages of network feature extraction. After the features are obtained, the features are sent to the classifier through the GAP and the fully connected layer for prediction to obtain the prediction result of the training set and finally use the optimizer to update the gradient and trainable parameters in the network by minimizing the cross-entropy loss function. The cross-entropy loss is defined as

$$\text{Loss} = -\frac{1}{M}\sum_{m=1}^{M}\left(y^{(m)}\right)^T \log p^{(m)} \tag{13}$$

where M represents the number of samples and $y^{(m)} \in R^C$ represents the one-hot vector of the real class corresponding to the mth image patch in the training set, and $p^{(m)} \in R^C$ represents the class prediction probability of the mth image patch in the training set. By continuously iterating and minimizing the loss of cross entropy, the predicted labels are constantly approaching the distribution of real labels.

After the network training, input the test set X_{test} and Y_{test} into the trained model and get the prediction result $\hat{Y}_{\text{test}}$. According to the prediction result, use the evaluation index to verify the effectiveness of the model. Finally, input X_l into the model to get the prediction labels of these image patches and draw the corresponding classification map according to the prediction labels.

The specific implementation details of the training and testing of the method proposed in this article are shown in Algorithm 1.

III. Experimental Results

A. Dataset Description

1) IP Dataset: This dataset contains 16 labeled land cover types. The size of the original HSI is 145 × 145, the spatial resolution of each pixel is 20 m/pixel, and it contains 224 spectral bands ranging from 0.4 to 2.5 μm. In the experiment, the remaining 200 bands after removing the water absorption bands and the blank in the original HSI were taken. In addition, 5% of the labeled pixels are selected as the training set to train the model, and the rest are used as the test set to verify the performance of the model. The detailed descriptions of the training samples and test samples are shown in Table I. Fig. 4(a) and (b) shows the false-color image and the ground truth map, respectively.

2) Houston Dataset: This dataset has 15 land cover types labeled in the scene. The HSI contains 144 bands with a wavelength range of 0.364–1.046 μm. It contains 1905 × 349 pixels with a spatial resolution of 2.5 m/pixel. In the experiment, 5% labeled samples from each class are selected as the training set and the rest as the test set. The detailed description of the number of training and test samples is shown in Table II, and the false-color map and ground truth map are shown in Fig. 5(a) and (b), respectively.

3) PC Dataset: This dataset has nine land cover types in the scene. The HSI contains 102 bands and 1096 × 715 pixels with a spatial resolution of 1.3 m/pixel. In the experiment, 0.3% labeled samples from each class are selected as the training set and the rest as the test set. The detailed description of the number of training and test samples of each class is

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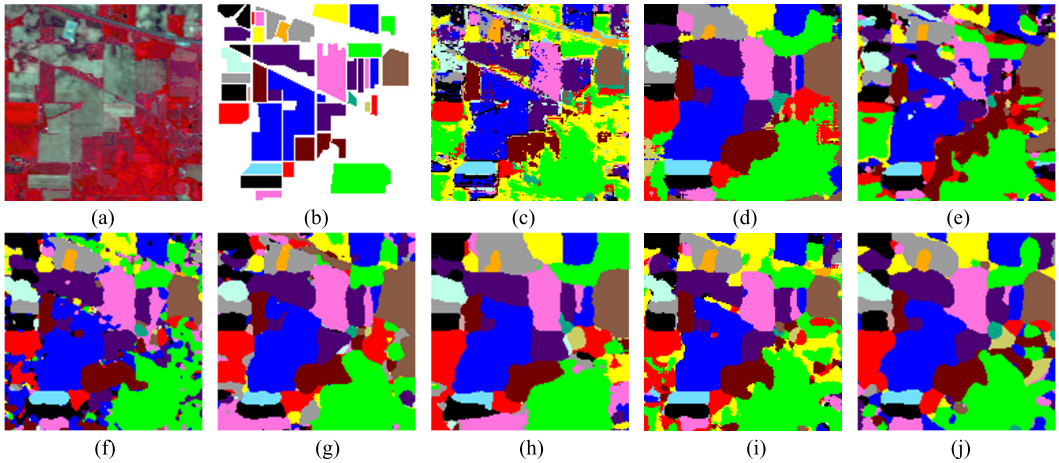


Fig. 4. Classification maps obtained by different classification methods on IP dataset. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) DFFN. (e) 2-D CNN. (f) 3-D CNN. (g) HybridSN. (h) MCNN-CP. (i) CEGCN. (j) Proposed method.

TABLE I

NUMBER OF TRAINING SAMPLES (5%) AND TEST SAMPLES FOR IP DATASET

No.	Color	Class Name	Train	Test	Labeled
1		Alfalfa	3	43	46
2		Corn-notill	72	1356	1428
3		Corn-mintill	42	788	830
4		Corn	12	225	237
5		Grass-pasture	25	458	483
6		Grass-trees	37	693	730
7		Grass-pasture-mowed	2	26	28
8		Hay-windrowed	24	454	478
9		Oats	1	19	20
10		Soybean-notill	49	923	972
11		Soybean-mintill	123	2332	2455
12		Soybean-clean	30	563	593
13		Wheat	11	194	205
14		Woods	64	1201	1265
15		Buildings-Grass	20	366	386
16		Stone-Steel-Towers	5	88	93
		Total Samples	520	9729	10249

TABLE II

NUMBER OF TRAINING SAMPLES (5%) AND TEST SAMPLES FOR HOUSTON DATASET

No.	Color	Class Name	Train	Test	Labeled
1		Healthy grass	69	1305	1374
2		Stressed grass	73	1381	1454
3		Synthetic grass	40	755	795
4		Trees	63	1201	1264
5		Soil	65	1233	1298
6		Water	17	322	339
7		Residential	74	1402	1476
8		Commercial	68	1286	1354
9		Road	78	1476	1554
10		Highway	71	1353	1424
11		Railway	78	1488	1566
12		Parking Lot1	71	1358	1429
13		Parking Lot2	32	600	632
14		Tennis court	26	487	513
15		Running track	40	758	798
		Total Samples	865	16405	17270

shown in Table III, and the false-color map and ground truth map are shown in Fig. 6(a) and (b), respectively.

TABLE III

NUMBER OF TRAINING SAMPLES (0.3%) AND TEST SAMPLES FOR PC DATASET

No.	Color	Class Name	Train	Test	Labeled
1		Water	198	65773	65971
2		Trees	23	7575	7598
3		Asphalt	9	3081	3090
4		Self-Blocking Bricks	8	2677	2685
5		Bitumenh	20	6564	6584
6		Tiles	28	9220	9248
7		Shadows	22	7265	7287
8		Meadows	128	42698	42826
9		Bare Soil	9	2854	2863
		Total Samples	445	147707	148152

TABLE IV

NUMBER OF TRAINING SAMPLES (3%) AND TEST SAMPLES FOR KSC DATASET

No.	Color	Class Name	Train	Test	Labeled
1		Scrub	23	738	761
2		Willow swamp	8	235	243
3		CP hammock	8	248	256
4		Slash pine	8	244	252
5		Oak/Broadleaf	5	156	161
6		Hardwood	7	222	229
7		Grass-pasture-mowed	4	101	105
8		Graminoid marsh	13	418	431
9		Spartina marsh	16	504	520
10		Cattail marsh	13	391	404
11		Salt marsh	13	406	419
12		Mud flats	16	487	503
13		Water	28	899	927
		Total Samples	162	5049	5211

4) KSC Dataset: This dataset contains 13 labeled land cover types. The dataset has 512 × 614 pixels with a spatial resolution of 18 m/pixel and 224 spectral bands with a wavelength range of 0.4–2.5 μm. After removing the water absorption bands and low signal-to-noise ratio bands, the remaining 176 bands are used as the experimental data. Select 3% of the labeled samples from each class as the training set and the rest as the test set. The detailed number of labeled data for each class is shown in Table IV, and the false-color

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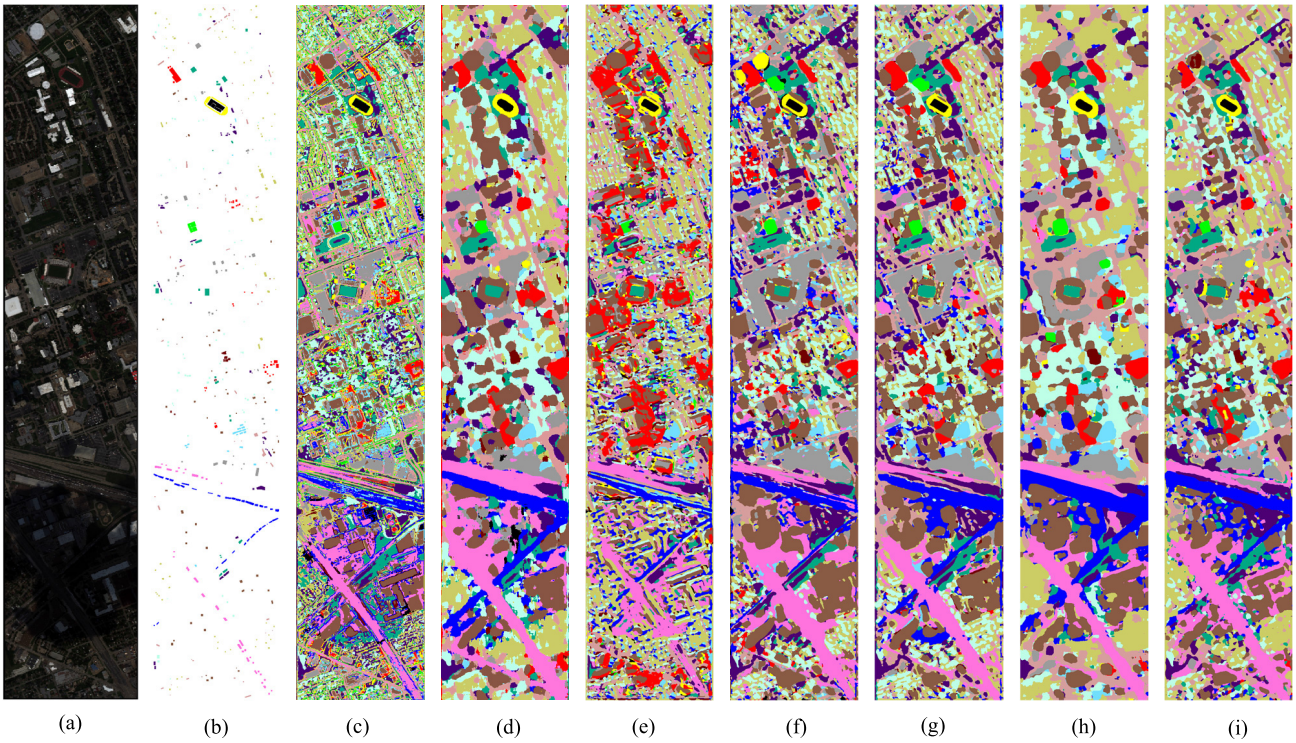


Fig. 5. Classification maps obtained by different classification methods on Houston dataset. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) DFFN. (e) 2-D CNN. (f) 3-D CNN. (g) HybridSN. (h) MCNN-CP. (i) Proposed method.

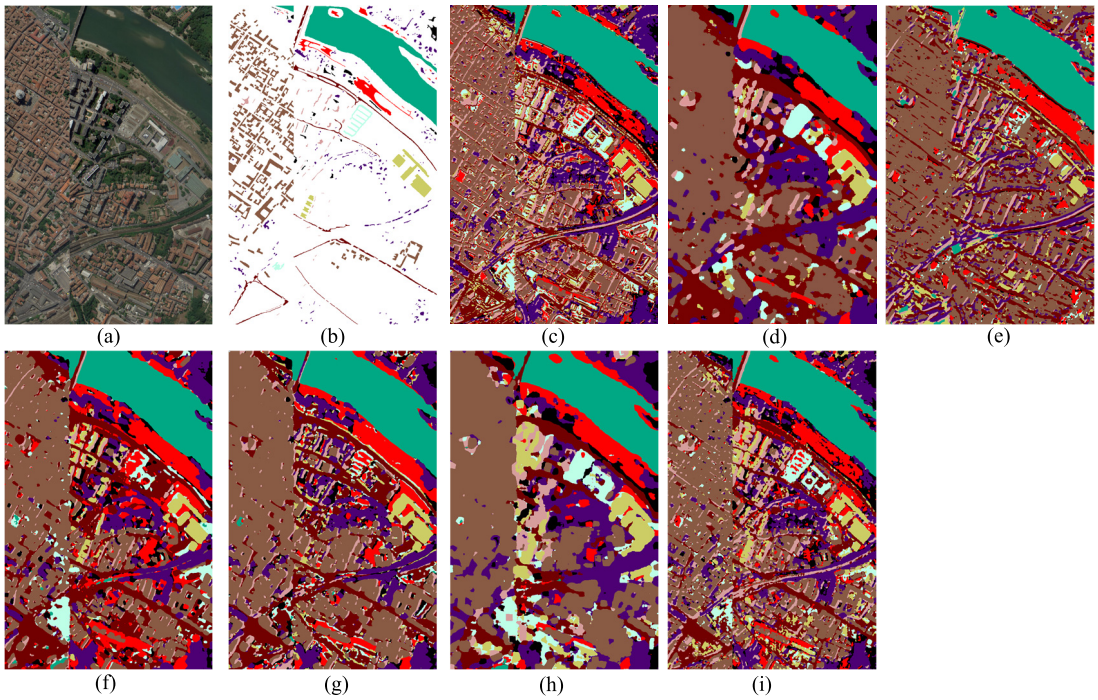


Fig. 6. Classification maps obtained by different classification methods on PC dataset. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) DFFN. (e) 2-D CNN. (f) 3-D CNN. (g) HybridSN. (h) MCNN-CP. (i) Proposed method.

map and ground truth map are shown in Fig. 7(a) and (b), respectively.

B. Experimental Setup

The spatial sizes of input image patches of the proposed method are selected as 15×15, 17×17, 15×15, and 21×21 on the four datasets, respectively. The number of channels of the convolutional layer in each feature extraction stage of the network is set to 32, 64, and 128, respectively, and the 1-D convolution kernel size k is set to 5 in the SOCA of each stage. In the experiment, Adam is used as the optimizer to optimize the parameters of the network, and the learning

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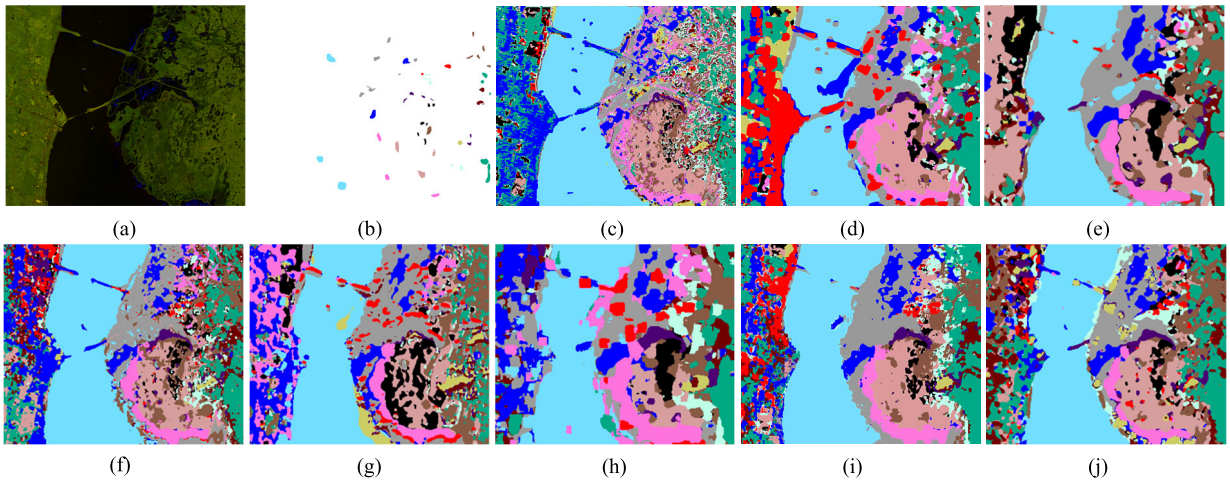


Fig. 7. Classification maps obtained by different classification methods on KSC dataset. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) DFFN. (e) 2-D CNN. (f) 3-D CNN. (g) HybridSN. (h) MCNN-CP. (i) CEGCN. (j) Proposed method.

rate is set to 0.001. The batch size and training epochs are 32 and 100, respectively. The proposed model is implemented by tensorflow framework. The experimental environment is Intel[1] Xeon[1] CPU, a 25G memory, and a Tesla P100 GPU, and the operating system is Ubuntu 18.04.

In the experiment, three commonly used evaluation indicators are used to verify the effectiveness of the proposed method, namely, overall accuracy (OA), average accuracy (AA), and Kappa coefficient. In order to reduce the randomness of the experimental results, each dataset runs ten times, and the mean and variance of the ten experimental results are obtained as the basis for evaluating the performance of all classification methods.

C. Comparison Methods

In order to verify the effectiveness of the proposed method in HSI classification, this article compares with seven advanced methods on four public hyperspectral datasets [IP, Houston, PC, Kennedy Space Center (KSC)]. The seven classification methods are: contextual deep CNN (CDCNN) [44], DFFN [40], 2-D CNN [26], 3-D CNN [54], HybridSN [37], mixed convolutions and covariance pooling (MCNN-CP) [50], and CNN-enhanced graph convolutional network (CEGCN) [55]. CDCNN uses a convolution kernel group of different sizes to extract multiscale information at the head of the network. DFFN fuses the complementary information of different hierarchical layers. 2-D CNN performs PCA transformation on the original HSI and sends the first principal component into the network to extract spatial features using 2-D convolution; 3-D CNN first performs PCA on the original HSI and extracts the first 20 principal components. Then, 3-D convolution kernel is used to extract spectral and spatial features. HybridSN combines 2-D convolution and 3-D convolution. First, 3-D convolution is used to extract spectral spatial features, and then, 2-D convolution is used to further extract more abstract spatial features. MCNN-CP uses a combination of 3-D convolution and 2-D convolution to learn spatial spectral features and then uses covariance pooling at the end of the network to extract the second-order features of the image. CEGCN uses CNN to extract the spatial features in the fixed window and uses graph convolution network to extract node features of arbitrary shape. Because Houston and PC dataset have a large number of pixels, when training CEGCN on these two datasets, a large number of nodes will be generated, resulting in the occupied video memory exceeding the capacity of the experimental equipment. Therefore, this method is only used for comparison on IP and KSC datasets.

D. Classification Results of Different Methods

Tables V–VIII show the numerical classification results of the proposed method and the comparison methods on four datasets, namely, OA, AA, Kappa coefficient, classification accuracy of each class, training time, and test time. Bold fonts in all tables represent the best classification performance of all methods. Figs. 4–7 show the classification maps of the proposed method and comparison methods on four datasets. Table V lists the classification results of the proposed method and the comparison methods on IP dataset.

Table V shows that compared with the comparison methods, MCRSCA has the best classification accuracy in six of the total 16 classes, and the OA, AA, and Kappa coefficients of the proposed method are all the highest. In addition to the methods proposed in this article, the best method of classification performance is CEGCN and the worst is CDCNN. This is mainly because CEGCN can extract local spatial information using CNN and extract global node information of arbitrary shape using graph convolution network. However, the size of input image patches of CDCNN is too small to extract limited spatial context information. In addition, there are only two and one training samples for class 7 and class 9, respectively. Except MCRSCA, all other methods have poor classification performance in one of these two categories. The OA of the comparison methods is from 17.04% to 92.48% in class 7 and from 10.53% to 75.59% in class 9. Although the OA result of CEGCN is similar to that of MCRSCA, the accuracy of CEGCN in class 9 is only 22.94%.

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TABLE V
CLASSIFICATION RESULTS OF CDCNN, DFFN, 2-D CNN, 3-D CNN, HYBRIDSN, MCNN-CP, CEGCN, AND MCRSCA ON IP DATASET

Class	CDCNN	DFFN	2DCNN	3DCNN	HybridSN	MCNN-CP	CEGCN	MCRSCA
1	28.14	92.09	54.09	73.05	87.27	89.09	72.83	**98.18**
2	77.46	95.55	87.35	94.65	92.93	94.89	**96.99**	96.29
3	64.51	94.63	84.01	93.15	90.61	95.46	**97.03**	96.92
4	50.13	92.22	84.00	92.32	84.62	**97.33**	92.12	96.76
5	82.97	92.49	74.90	92.59	93.77	95.16	95.10	**96.69**
6	96.25	96.35	91.00	96.72	96.05	93.28	**99.56**	97.36
7	36.15	82.31	45.19	33.33	54.07	77.04	92.48	**92.96**
8	98.59	99.78	98.90	99.50	99.65	99.82	**99.85**	99.82
9	32.63	41.58	38.95	40.60	57.89	75.79	22.94	**80.53**
10	72.33	93.22	80.82	92.71	93.76	95.41	95.50	**96.14**
11	80.96	96.99	92.51	95.30	96.74	96.81	**98.47**	97.45
12	53.45	89.61	81.28	92.46	85.86	93.68	**97.35**	96.18
13	97.22	97.22	92.00	99.41	94.67	91.59	**99.45**	97.13
14	95.15	99.21	96.36	98.13	97.94	97.95	**99.54**	99.37
15	72.43	96.50	82.40	90.83	85.56	97.06	93.08	**99.00**
16	96.14	77.84	70.68	96.75	73.64	82.27	**95.60**	94.10
OA (%)	79.44±0.98	95.51±0.41	87.99±1.27	94.63±0.84	93.71±1.02	95.73±0.62	97.28±0.44	**97.32±0.38**
AA (%)	70.91±2.76	89.87±1.95	78.40±2.43	86.35±2.90	86.56±2.74	92.04±1.81	90.49±2.43	**95.93±1.56**
Kappa (×100)	76.51±1.11	84.87±0.47	86.25±1.47	93.86±0.96	92.82±1.58	95.13±0.71	96.90±0.51	**96.94±0.43**
Train time (s)	19.75	61.56	21.25	51.53	35.63	105.20	**11.26**	41.75
Test time (s)	3.57	6.11	0.77	2.41	1.71	4.78	**0.51**	2.78

TABLE VI
CLASSIFICATION RESULTS OF CDCNN, DFFN, 2-D CNN, 3-D CNN, HYBRIDSN, MCNN-CP, AND MCRSCA ON HOUSTON DATASET

class	CDCNN	DFFN	2DCNN	3DCNN	HybridSN	MCNN-CP	MCRSCA
1	96.97	96.80	85.63	96.74	96.89	96.26	**97.85**
2	92.11	97.97	72.98	**99.43**	99.28	97.54	99.37
3	96.11	98.01	97.35	99.03	99.23	97.88	**99.31**
4	93.54	**92.97**	72.11	94.86	95.62	**90.18**	**97.87**
5	99.11	99.43	92.63	99.90	99.91	99.75	**100.00**
6	88.82	93.20	66.99	92.42	92.98	91.77	**94.19**
7	89.22	**95.82**	87.88	91.11	93.12	92.63	95.81
8	84.81	93.94	67.15	91.95	92.30	92.62	**94.18**
9	85.89	**94.80**	89.18	90.38	93.55	93.96	94.47
10	91.87	99.28	91.18	98.87	99.44	**100.00**	99.50
11	89.46	**99.44**	95.22	98.23	99.11	98.49	98.76
12	93.66	98.08	73.55	98.13	**98.27**	97.78	98.20
13	92.75	94.70	94.35	93.37	94.30	96.13	**96.20**
14	94.74	99.96	96.57	**100.00**	**100.00**	**100.00**	99.98
15	98.47	98.40	94.18	**99.83**	99.51	97.16	99.58
OA (%)	92.13±0.91	96.94±0.32	84.70±0.91	96.20±0.61	96.92±0.54	96.15±0.81	**97.70±0.50**
AA (%)	92.50±0.78	96.85±0.33	85.13±0.96	96.28±0.65	96.90±0.45	96.14±0.73	**97.68±0.56**
Kappa (×100)	91.49±0.98	96.71±0.33	83.46±0.99	95.89±0.66	96.67±0.59	95.84±0.88	**97.51±0.54**
Train time (s)	94.17	153.33	**14.22**	63.31	61.79	80.89	67.54
Test time (s)	8.11	13.94	**1.36**	3.61	3.42	4.40	4.41

However, the OA of the MCRSCA is 92.96% and 80.53% in class 7 and class 9, respectively, indicating that the comparison methods may ignore the class with small size of samples and only focus on the class with large size of samples. However, the proposed method can obtain good classification accuracy even in the classes with class unbalanced and few samples. In terms of training and test time, CEGCN has the least training and testing time. This is because CEGCN uses superpixel segmentation method to reduce the number of graph nodes and has fewer training parameters. The method with the longest training time is MCNN-CP, and the training time of MCRSCA is in the middle among all methods. Fig. 4 shows the classification maps of multiple comparison methods on IP dataset. As can be seen from Fig. 4, the classification map of CDCNN is filled with isolated noise, resulting in a large number of other classes being distributed in homogeneous areas. This is mainly because this method only uses a group of multiscale convolution in the first layer of the network, and the spatial size of the input image patches is too small, so this method cannot effectively extract more abundant spatial context information, resulting in inconsistent spatial distribution of the classification map. Other comparison methods use larger image patches as input and have better visual performance than those methods with small size of image patches as input. However, in the classification maps of 2-D CNN and 3-D CNN, the labels of homogeneous regions are not smooth, and there are still large areas of other classes of objects in homogeneous regions. Although the

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TABLE VII
CLASSIFICATION RESULTS OF CDCNN, DFFN, 2-D CNN, 3-D CNN, HYBRIDSN, MCNN-CP, AND MCRSCA ON PC DATASET

Class	CDCNN	DFFN	2DCNN	3DCNN	HybridSN	MCNN-CP	MCRSCA
1	99.89	99.99	99.88	99.99	**100.00**	99.91	**100.00**
2	94.44	94.09	33.93	91.87	85.43	89.09	**95.84**
3	86.28	83.64	15.02	81.89	49.76	76.48	**88.17**
4	77.81	**95.99**	34.36	73.03	48.21	88.90	86.93
5	92.35	**96.54**	79.70	92.59	88.79	92.92	94.59
6	96.04	97.10	83.50	96.68	89.55	86.85	**98.34**
7	87.83	94.93	71.63	94.71	89.87	**96.18**	95.67
8	99.53	99.29	94.68	99.60	99.72	99.07	**99.68**
9	**99.39**	81.31	73.20	72.01	65.54	61.84	92.38
OA (%)	97.64±0.21	98.13±0.19	88.21 ± 0.63	97.26±0.44	94.87±1.56	96.38±0.43	**98.51±0.32**
AA (%)	92.62±1.22	93.65±0.67	65.10±1.95	89.15±1.47	79.65±5.10	87.92±1.55	**94.62±1.22**
Kappa (×100)	96.67±0.30	97.35±0.27	83.03±0.92	96.11±0.64	92.70±2.23	94.85±0.61	**97.88±0.45**
Train time	35.22	65.69	**10.71**	33.42	53.50	65.97	41.43
Test time	39.81	67.55	**8.94**	21.12	46.32	64.94	30.72

classification maps of HybridSN, MCNN-CP, and CEGCN have good spatial consistency, the classification accuracy is lower than the method proposed in this article.

Table VI lists the classification performance of different methods on Houston dataset.

Table VI shows that MCRSCA obtains the best classification results among the three evaluation indicators of OA, AA, and Kappa. Compared with the other classification methods, MCRSCA has the highest classification accuracy in seven categories. The two methods with the highest classification accuracy among other methods are DFFN and HybridSN. This is because DFFN fuses the residual blocks of different layers to obtain more complementary information and HybridSN uses 3-D and 2-D convolution to extract rich spectral and spatial information, but OA of MCRSCA is still 0.76%, 0.78% higher than these two methods, respectively. The method with the worst classification performance is 2-D CNN. The main reason is that this method only uses the first principal component of the original spectrum as the input. Although it reduces the training time, it inevitably loses some useful information, resulting in the decline of classification performance. In terms of training time, 2-D CNN takes the least time, while DFFN takes the most time. The method proposed in this article is located between 3-D CNN and MCNN-CP, but it achieves higher accuracy with a medium training time. Fig. 5 shows the classification maps of different classification methods on Houston dataset.

Fig. 5 shows that small granular noise exists in the classification map of CDCNN, which seriously affects its classification accuracy and visual performance. Because 2-D CNN takes large spatial neighborhood pixels as an input, it is blurred in the edge part and thus interferes with the correct classification of center pixels; 3-D CNN has a good performance in keeping the edges correctly divided, but there is much small granular noise in the classification map. The visual classification performance of HybridSN and DFFN is good, but these two methods take large image patches as an input, and there is still a phenomenon of over smoothing in their classification map. Especially for MCNN-CP, which has a large area of the same class, resulting in rough and fuzzy edge division. The method proposed in this article can extract more detailed multiscale features of HSI, so the classification of edge parts is more accurate while maintaining the spatial consistent distribution of ground objects.

Table VII shows the classification results of different classification methods on PC datasets.

As can be seen from Table VII, in all nine classes, the classification accuracy of the proposed method is better than that of the comparison methods in five classes, and the OA, AA, and Kappa coefficients are all optimal. The OA of MCRSCA is 0.38%–10.30% higher than that of all comparison methods. The comparison methods with the best and the worst classification results are DFFN and 2-D CNN, respectively. MCRSCA achieves the best classification results in the case of medium time-consuming, which proves the effectiveness of this method. Fig. 6 shows the classification maps of the proposed method and the comparison methods on the PC dataset.

It can be seen from Fig. 6 that the classification maps of DFFN, HybridSN, and MCNN-CP are over smooth because these methods take a large size of image patches as the input, so that most pixels of the same category in the image patches interfere with the classification of the central pixel. Although 2-D CNN classifies smoothly in the boundary area, its classification map has a large number of misclassified areas in the homogeneous area, which makes it unable to maintain spatial consistency. CDCNN, 3-D CNN, and MCRSCA all have good visual classification performance, but there is fuzzy boundary division in the classification map of 3-D CNN. CDCNN divides relatively flat areas at the boundary, but there are misclassified areas. The classification map of MCRSCA can jointly maintain boundary flatness and spatial consistency and achieves good visual results in maintaining correct labels and spatial consistency in the boundary area.

Table VIII lists the classification performance of different classification methods on the KSC dataset.

As can be seen from Table VIII, compared with the comparison methods, OA, AA, and Kappa coefficient of the proposed method are all the highest. The proposed method achieves the best classification results in 11 classes out of a total of 13 classes, and the classification accuracy of six classes is 100.00%. The methods whose classification

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TABLE VIII
CLASSIFICATION RESULTS OF CDCNN, DFFN, 2-D CNN, 3-D CNN, HYBRIDSN, MCNN-CP, CEGCN, AND THE PROPOSED METHOD ON KSC DATASET

Class	CDCNN	DFFN	2DCNN	3DCNN	HybridSN	MCNN-CP	CEGCN	MCRSCA
1	95.38	98.71	99.70	99.59	99.16	99.32	99.31	**99.97**
2	78.17	76.04	85.08	97.25	74.84	99.15	93.92	**99.92**
3	84.76	95.04	96.93	98.19	81.78	99.43	99.03	**99.84**
4	46.47	72.83	82.62	78.16	69.75	95.74	77.96	**96.89**
5	51.22	72.11	86.54	82.44	76.54	**88.85**	79.87	**88.85**
6	65.86	84.41	96.85	93.20	84.00	98.56	**99.45**	99.28
7	92.28	98.71	98.24	89.22	91.57	**100.00**	95.84	99.61
8	75.89	92.78	97.18	99.04	96.08	98.33	**99.81**	**100.00**
9	93.83	97.48	**100.00**	97.92	98.43	**100.00**	**100.00**	**100.00**
10	82.81	98.21	**100.00**	**100.00**	99.97	**100.00**	99.95	**100.00**
11	97.41	99.61	**100.00**	99.56	99.78	99.56	99.97	**100.00**
12	88.97	96.22	**100.00**	97.87	99.24	94.34	98.88	**100.00**
13	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA (%)	86.60±2.08	94.18±3.05	97.48±0.68	97.08±0.86	94.18±1.02	98.50±0.54	97.64±0.91	**99.44±0.48**
AA (%)	81.00±2.63	90.94±4.74	95.63±1.00	94.80±1.71	90.08±1.81	97.94±0.60	95.69±1.72	**98.80±1.01**
Kappa (×100)	85.07±2.32	93.53±3.38	97.16±0.75	96.75±0.96	93.51±1.14	98.33±0.60	97.37±1.01	**99.38±0.54**
Train time	**11.98**	27.95	14.00	14.90	72.88	99.06	301.46	17.96
Test time	1.75	3.63	**1.48**	1.93	6.60	9.44	7.56	2.01

accuracy is close to the proposed method are MCNN-CP, 2-D CNN, and 3-D CNN, but the OA of the proposed method is still 0.95%, 2.28%, and 2.68% higher than those of the three methods. In addition, CDCNN adopted small size of patches as input and the OA of CDCNN did not exceed 90%, while the classification accuracy of 2-D CNN in the 9th–13th categories reached 100.00% and the OA is 97.48%, which further illustrates the importance of spatial information for classification. In terms of training time, the two most time-consuming methods are CEGCN and MCNN-CP. The training time exceeds 300 and 90 s, respectively, while MCRSCA takes less than 18 s.

Fig. 7 shows the classification maps of different methods on the KSC dataset.

As can be seen from Fig. 7, there is still a lot of noise of particle shape in the classification map of CDCNN. Since 2-D CNN and MCNN-CP use a large spatial size of neighborhood pixels as the input of the model, the classification map is too smooth, and a large number of regions of other categories are generated in the classification map. There is a large area of misclassification in the classification map of DFFN. Although 3-D CNN and HybridSN have better visual performance, these two methods for Slash pine and Oak/Broadleaf target recognition have poor classification results. The proposed MCRSCA has the highest classification accuracy while maintaining spatial consistency of ground objects.

E. Influence of the Spatial Size on the Result

The spatial size of the input image patches is an important factor affecting the performance of classification. If the spatial size is too small, rich spatial information cannot be fully utilized, while if the spatial size is too large, there may be interference information generated by samples of other classes in the adjacent pixels, leading to misclassification. Therefore, in order to verify the influence of different spatial sizes of input image patches on classification results, 11 × 11, 13 × 13, 15 × 15, 17 × 17, 19 × 19, and 21 × 21 are used as input spatial sizes of the proposed methods on four datasets. The classification results are shown in Table IX.

TABLE IX
OA (%) OF IMAGE PATCHES WITH DIFFERENT SPATIAL SIZES ON FOUR DATASETS

Spatial Size	IP	Houston	PC	KSC
11×11	96.00	96.52	98.04	97.64
13×13	97.01	97.30	98.22	98.32
15×15	**97.32**	97.55	**98.51**	98.83
17×17	97.09	**97.70**	98.25	99.23
19×19	97.01	97.41	98.07	99.25
21×21	96.73	97.24	98.06	**99.44**

It can be seen from Table IX that the classification performance on the IP dataset increases first and then decreases with the increase of the spatial size. When the input size reaches 15 × 15, the best classification performance is achieved. The PC dataset is similar to IP, and the best classification performance is obtained when the input spatial size is 15 × 15. The OA of Houston dataset increases with the increase of spatial size, when the spatial size is 17 × 17, OA reached the maximum, and OA begins to decline when the space size is further increased. The classification accuracy of KSC is continuously improved with the increase of the spatial size. When the spatial size is 21 × 21, the best classification performance is achieved. In the four datasets, the accuracy of the spatial size with the best classification performance is 1.32%, 1.18%, 0.47%, and 1.80% higher than that of the spatial size with the worst classification performance, respectively, which proves that the spatial size of the input image patches has a certain influence on the classification performance. From the pseudo color image of KSC dataset, the distribution of various ground objects is relatively regular, and there are large homogeneous areas. The spatial size of KSC dataset is the largest, and the spatial range of homogeneous areas is large. Therefore, using large spatial HSI patches on KSC dataset can obtain more abundant spatial information. From the pseudo color images of IP, Houston, and PC datasets, the spatial distribution of ground objects in these three datasets

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TABLE X
OA (%) UNDER DIFFERENT NUMBERS OF TRAINING SAMPLES

IP	3%	5%	7%	10%
	95.00	97.32	98.12	98.79
Houston	1%	3%	5%	7%
	87.17	95.38	97.70	98.75
PC	0.1%	0.3%	0.5%	0.7%
	95.71	98.51	98.81	99.31
KSC	1%	2%	3%	5%
	95.13	97.99	99.44	99.72

TABLE XI
OA (%) OF THE PROPOSED METHOD USING SOCA OR SE MODULE

Module	IP	Houston	PC	KSC
With SOCA	97.32	**97.70**	**98.51**	**99.44**
With SE-module	**97.33**	97.64	98.42	99.33

is irregular, and there are a variety of ground objects mixed and distributed in the same area. Other types of pixels may interfere with the classification results of the central pixels. Therefore, the spatial size of IP, Houston, and PC datasets should not be too large.

F. Performance Under Different Numbers of Training Samples

The labeled samples in HSI are limited. Therefore, in order to study the influence of different numbers of training samples on the classification results, experiments are carried out on four datasets. Training samples of 3%, 5%, 7%, and 10% on IP dataset, 1%, 3%, 5%, and 7% on Houston dataset, 0.1%, 0.3%, 0.5%, and 0.7% on PC dataset, and 1%, 2%, 3%, and 5% on KSC dataset are selected for the experiment. The experimental results are shown in Table X.

As can be seen from Table X, when a small number of training samples are selected on each dataset, the MCRSCA can still obtain good classification accuracy. When the training samples are further increased, the classification performance on the four datasets is improved, which proves that the proposed method has better generalization performance.

G. Compared With SE Module

Squeeze-and-Excitation (SE) module [56] is a common channel attention module, which is used to explicitly model the interdependence between feature channels and obtain the importance of each channel. It has been proven to be effective in many computer vision tasks. In order to verify the effectiveness of SOCA proposed in this article, SOCA is replaced by SE module for experiment. The experimental results are shown in Table XI.

It can be seen from Table XI that the classification performance of SE module is close to that of SOCA, but the classification performance of SOCA on the three datasets is slightly higher than that of SE module, which proves the effectiveness of SOCA. In addition, the trainable parameters of SOCA are only related to the size of 1-D convolution kernel, which has fewer parameters than SE module.

In addition, in order to study the separability of the features extracted by the proposed method and the SE module,

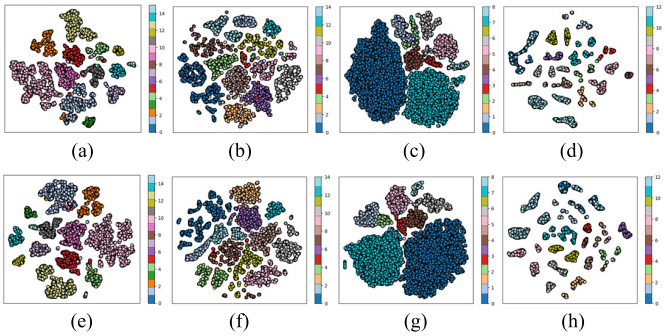


Fig. 8. Feature visualization via t-SNE on four datasets. (a)–(d) Feature visualization of SOCA. (e)–(h) Feature visualization of SE module.

TABLE XII
OA (%) OF MCBR AND CC ON FOUR DATASETS

Module	IP	Houston	PC	KSC
MCBR	**97.32**	**97.70**	**98.51**	**99.44**
CC	97.01	97.30	97.90	99.18

t-distributed Stochastic Neighbor Embedding (t-SNE) [57] algorithm is used to visualize the separability of the features extracted from the last layer of the network. The visualization of features is shown in Fig. 8.

In Fig. 8, the points of different colors represent different classes of samples. The smaller the distance between points of the same color and the larger the distance between the points of different colors, the better the separability of features. As can be seen from Fig. 8, the feature visualization using both the MCRSCA and the SE module has the smaller distance between the same class of samples and has an obvious spatial interval between different classes of samples. This proves that both the proposed method and SE module have good feature representation ability.

H. Compared With Content-Aware Convolution

Both content-aware convolution (CC) [58] and MCBR can extract rich multiscale spatial context information. In order to prove the effectiveness of MCBR, the experiment is conducted to compare with CC. The experimental results are shown in Table XII.

As can be seen from Table XII, the OA of MCBR on the four datasets is higher than that of CC, which proves that MCBR can effectively extract richer multiscale context information.

I. Ablation Experiment

In order to verify the contribution of the proposed modules to classification performance, image patches with spatial size of 15×15 are used as input for ablation experiments in four datasets, and the experimental results are shown in Table XIII.

In Table XIII, $\times$ represents that one of the modules is not used in each residual block, and $\surd$ represents that one of the modules is used in each residual block. As can be seen from the third row of Table XIII, when only MCBR is used, the classification performance of IP and KSC is

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TABLE XIII
IMPACT OF DIFFERENT MODULES ON OA (%)

MCBR	SOCA	IP	Houston	PC	KSC
×	×	96.58	97.35	98.44	97.92
√	×	96.90	97.29	98.32	98.60
×	√	96.76	**97.66**	98.50	98.80
√	√	**97.32**	97.55	**98.51**	**98.83**

TABLE XIV
OA (%) OF THE COMBINATION OF CONVOLUTION KERNELS WITH DIFFERENT SIZES

1×1	3×3	5×5	IP	Houston	PC	KSC
√	√	×	97.27	97.60	98.41	99.13
√	×	√	97.25	**97.78**	98.37	99.35
×	√	√	97.27	97.69	98.27	99.35
√	√	√	**97.32**	97.70	**98.51**	**99.44**

improved. As can be seen from the fourth row of Table XIII, when SOCA is only used, OA on the four datasets has been improved, indicating that embedding SOCA separately at each stage of the network is beneficial to the improvement of accuracy. As can be seen from the fifth line of Table XIII, when MCBR and SOCA are embedded simultaneously in each stage, OA increases by 0.74%, 0.07%, and 0.91% on IP, PC, and KSC datasets, respectively. According to this result, the accuracy of embedding two modules together is higher than that of embedding one module alone on these three datasets. After the two modules are embedded into the network together, on the one hand, more global spatial details are extracted; on the other hand, the correlation dependence between channels is obtained, and the representation ability of features is enhanced, so the classification results will be further improved when the two modules are added to each stage of feature extraction.

In order to study the influence of the combination of convolution kernels with different scales on the classification results, the combination of convolution kernels with different sizes is selected on MCRSCA and experiments are carried out on four HSI datasets. The experimental results are shown in Table XIV.

Since the spatial structures of ground objects are different, it can be seen from Table XIV that the classification results caused by the combination of convolution kernels of various sizes are different; 1×1 convolution kernel is used to capture the correlation between spectral channels, and 3×3 and 5×5 convolution kernels are used to extract multiscale spatial structure information. When these three convolution kernels are used together, the best classification performance is obtained on IP, PC, and KSC datasets. The combination of convolution kernels of 1×1 and 5×5 achieves the best classification performance on Houston dataset.

IV. CONCLUSION

In this article, a novel MCRSCA method is proposed for HSIC. To extract spatial context information and obtain second-order statistical information of features, MCBR and SOCA are embedded in MCRSCA. The MCBR integrates concatenation and addition to fuse features and transmits multiscale information in the network through cross-branch interaction. The SOCA uses the covariance matrix to obtain the correlation between the paired channels. The attention mechanism is used to assign weights to channels adaptively, which fully captures the long-range dependence between channels and enhances the nonlinear expression ability. By integrating these two modules into the residual block through a skip connection and stacking multiple stages in turn, the multilevel features are extracted, respectively. Four hyperspectral datasets are experimented with a small number of training samples, and the MCRSCA is compared with seven advanced methods. The experimental results show that MCRSCA outperforms the comparison methods on all datasets. In addition, from the perspective of the classification maps, the visual performance of the MCRSCA is relatively smooth, and the result of edge region is relatively flat, which proves that MCRSCA can effectively extract multiscale spatial details and refine the classification results. However, the proposed method does not integrate the complementary information of the features between different layers of the network. Future work will consider fusing the features between the layers of the network to further improve the classification performance. The training time and parameters of the proposed method are not the least. Therefore, it can also be improved in reducing the parameters and training time.

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